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Branding is what people say about you when you are not in the room.



-Jeff Bezos



cles has added to the semiconductor BoM in two-wheelers. Note that in higher end two-wheeler models, the semiconductor BoM is much higher as the vehicle has a sophisticated ECU, multiple sensors, etc. All of this has pushed up semiconductor consumption in the petrol two-wheeler segment. EVs have even more electronics and semiconductor components,” says Menon.

India, one the fastest developing countries in the world, has announced a production linked incentive scheme worth 181 billion rupees dedicated towards promoting ACC battery manufacturing in the country.

“From a market size perspective, India produced around 25 million two-wheelers in 2019. As we see more functionality getting added in terms of sensors, controllers, communication protocols, entertainment options, etc, the semiconductor and electronic BoM is going to sharply increase in the days ahead,” Menon explains.

The shares of NXP Semiconductors, one of the leading suppliers of automotive semiconductors and solutions, delivered an RoI of 88.78 percent during the last one year. The company’s market cap as of 28th June 2021 was around 56.3 billion dollars. The stock price for NXP rose from \$108.15 per share (June 2020) to \$204.17 per share (June 2021).

The US, China, and India

The increasing importance of semiconductors can easily be gauged from how the world’s superpowers as well as India are making efforts to enable a superior local semiconductor ecosystem in their respective geographies. For starters, the USA alone has approved \$52 billion in spending for domestic semiconductor manufacturing and research as an important step towards strengthening the country’s leadership in the industry.

“We are in a competition to win the 21st century, and the starting gun has gone off,” the US President Joe Biden has said in a statement recently.

He added, “As other countries continue to invest in their own research and development, we cannot risk falling behind. America must maintain its position as the most innovative and productive nation on Earth.”

Similarly, India, one of the fastest developing countries in the world, is also working on envisaging a plan to facilitate the semiconductor ecosystem in the country. It is expected that the government of India will approach setting up of a fabless cum ATMP/OSAT ecosystem soon by offering handsome incentives to global heavy-weights.

China, on the other hand, has already given more than 40 billion yuan (\$6.19bn) in combined funding during the first five months of 2021 to more than 100 semiconductor companies. According to a report by Research and Markets, the investments in new fabs or capacity expansion will exceed US\$160 billion in China over the coming 5-7 years. This is expected to drive an increase in China’s equipment spending to more than \$40 billion in 2025, with sixty percent of the investments going to memory fabs.

All semiconductor companies, whether fabless or fab, have been spending billions of dollars in researching and developing the latest avatar of semiconductors and processes that enable faster and efficient manufacturing. In a recent interaction with Sunil Banwari, Advantest Corporation, we learnt that the top semiconductor producers are exploring industry 4.0 manufacturing capabilities in order to combine them with semiconductor design and fabrication.

“The shortage is likely to continue at least till the middle of the next year. That definitely fuels this trend that you have unearthed. With AI and 5G picking up, I think the market is going to be good for them. What Covid has done is accentuated the peak of the semiconductor industry,” says Hemant.

While the semiconductor industry is cyclical in nature, at the moment it seems far away from witnessing a down curve, primarily because the demand for modern electronics has just started going up. For investors it might be a good idea to invest in shares of such companies, but for governments it would be a good idea to go all guns blazing towards creating a superior semiconductor ecosystem. **EFY**

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